Integrated Experiments – Practical Report

1. Introduction

This practical demonstrates supervised classification using a Decision Tree and reinforcement learning using Q-Learning.

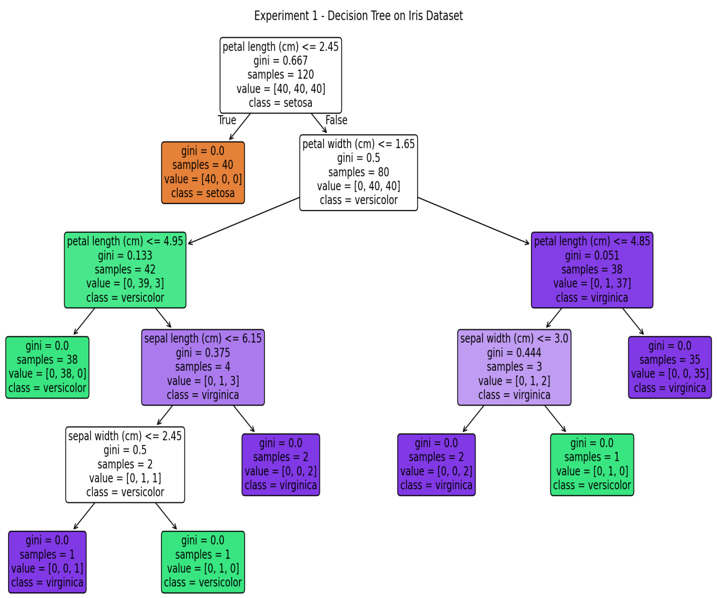
2. Experiment 1 – Decision Tree

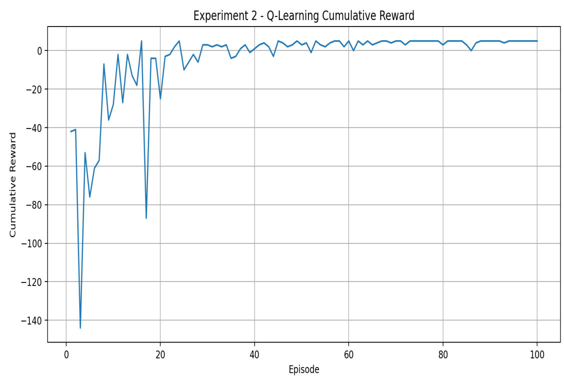
The Iris dataset is inspected, split into training and testing sets, and classified using a Decision Tree with the Gini criterion. Accuracy, confusion matrix and classification report are used for evaluation. Iris is a relatively well-separated classification problem, so high accuracy is expected.

3. Experiment 2 – Q-Learning

The agent learns navigation in a 4×4 Grid World through repeated interaction. Epsilon-greedy exploration initially encourages exploration and gradually shifts toward exploitation. Q-values become more informative as rewards are propagated backward from the goal.

4. Generated Output





5. Conclusion

The Decision Tree successfully learns class boundaries from labelled Iris samples. Q-Learning learns action values through trial and error and produces a policy for reaching the goal while avoiding obstacles and unnecessary movement.